SCHEMATIC - every wire of the set, grouped by job; neat, not to scale

Tables are pinouts (pin | signal) of the pins each panel uses; the PINOUT TABLES sheet has every pin. Wire label: id AWG colour [ident bands]. Dashed = built with the harness named in its table; grey header = an end on another harness.

WIRE COLOURS (R209): RED · BLK · ORN · BLU · WHT · GRN · YEL — **IDENT BANDS (R57):** Blue · Yellow · White — dashed = built with another harness

THROUGH THE BULKHEAD - every wire that lands on the DRB102 (signals, sensor rails, feeds)

ECU-A - Link G4X plug A (centre dash)
6-1437290-1

1	Injector 4 drive (-)	w111_c
2	Injector 3 drive (-)	w110_c
3	Injector 2 drive (-)	w109_c
4	Injector 1 drive (-)	w108_c
8	Crank sensor signal	w200_c
10	Coil 4 trigger (IGT)	w123_c
11	Coil 3 trigger (IGT)	w122_c
12	Coil 2 trigger (IGT)	w121_c
13	Coil 1 trigger (IGT)	w120_c
15	Coolant temp signal	w201_c
16	Intake temp signal	w202_c
17	MAP sensor signal	w206_c
21	Boost solenoid (-)	w103_c
22	Throttle pos 1 main	w128_c
27	EHPS pump speed	w_mrs_c
30	Turbo speed signal	w16_c
31	Flex fuel signal	w48_c
33	Throttle pos 2 sub	w129_c

ECU-B (Link G4X plug B, centre dash)
6-1437290-3

6	Oil temp signal	w203_c
7	Charge temp signal	w204_c
9	Knock sensor signal	w205_c
15	Oil pressure signal	w35_c
16	Fuel pressure signal	w32_c
18	ETB motor +	w124_c
23	Coolant press signal	w35_c
26	ETB motor -	w215_c
29	Gearbox speed signal	w_vss_sig_c

BUS-5V - DT06-12 plug (ECU splice point)
DT06-12SA

2	+5 V throttle sensor	w_rail_sp_5v_c
3	+5 V MAP sensor	w_5v_a_c39
4	+5 V turbo speed	w_5v_a_c40
5	+5 V oil pressure	w_5v_brail_c45
6	+5 V fuel pressure	w_5v_brail_c46
7	+5 V coolant press	w_5v_brail_c47

BUS-GNDOUT-A - DT06-12 plug (ECU splice point)
DT06-12SA

2	Gnd IAT/TPS/coolant	w_rail_sp_gndout_c
3	Gnd crank sensor	w_gnd_a_c41
4	Gnd cam sensor	w_gnd_a_c42
5	Gnd turbo speed	w_gnd_a_c43
6	Gnd flex + MAP	w_gnd_a_c44

BUS-GNDOUT-B - DT06-12 plug (ECU splice point)
DT06-12SA

6	Knock sensor (-)	w12_c
7	Gnd oil/crit/temps	w_gnd_b_c19
8	Gnd fuel press + VSS	w_gnd_b_c21

SP-8V - ECU splice point - SP-8V: +8 V cam sensor supply (E...
solder splice (node)

-	+8 V to cam sensor	w_rail_sp_8v_c

SP-CAM - At the ECU plugs - SP-CAM: cam signal + 1.8k pull...
solder splice (node)

-	Cam sensor signal	w_cam_sig_c

SP-CANH - CAN H join - ECU, CSB3, cluster, Pi, bulkhead (o...
solder splice (node)

-	CAN H to CAN-Lambda	wc_h_bh_bc

SP-CANL - CAN L join - ECU, CSB3, cluster, Pi, bulkhead (o...
solder splice (node)

-	CAN L to CAN-Lambda	wc_l_bh_bc

SP-REV - reverse switch out to CSB3 opto + reverse relay
solder splice (node)

-	Reverse sw +12 V out	w_rev_sw_c

SP-REV-GAUGE - GAUGE 15A splice - J/B 3 feed to reverse r...
solder splice (node)

-	Back-up sw +12 V in	w_rev_feed_c

SP-SW12 - ECU splice point - SP-SW12: switched +12 V (LO...
solder splice (node)

-	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	w_rail_sp_sw12_c
-	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	w_rail_sw12_b_c_ac

K-EPS - MR-S EHPS pump relay (HCR150, glove box, beside ...
925710-K000-0000

85	EHPS relay coil (-)	w_eps_rly

LOGIC box - custom fuse + relay box, glove box inboard (ca...
925710-K000-0000 · loom relay box

BUS2-1	EFI relay k_efi -> LOGIC LAM 7.5 A fuse	w_lam12_cp
BUS2-12	EFI relay k_efi -> LOGIC MRS 7.5 A fuse	w_fb_mrs_en
BUS2-3	EFI relay k_efi -> LOGIC COP 15 A fuse	w_cop_pwr_ac
BUS2-4	EFI relay k_efi -> LOGIC INJ 15 A fuse	w_inj_pwr_ac

POWER-NEG-BAR - POWER box NEG bar M8 stud (glove box)
stud

-	CAN-Lambda ground	w_lamg_ac

DRB102-CABIN - DRB102 receptacle (RH firewall)
DRB12-1025NE-4169

1	EFI relay k_efi -> LOGIC COP 15 A fuse	w_cop_pwr_ac
2	EFI relay k_efi -> LOGIC INJ 15 A fuse	w_inj_pwr_ac
9	Coil 1 trigger (IGT)	w120_c
10	Coil 2 trigger (IGT)	w121_c
11	Coil 3 trigger (IGT)	w122_c
12	Coil 4 trigger (IGT)	w123_c
13	+5 V fuel pressure	w_5v_brail_c46
14	Fuel pressure signal	w32_c
15	Gnd fuel press + VSS	w_gnd_b_c21
16	+5 V MAP sensor	w_5v_a_c39
17	MAP sensor signal	w206_c
18	Boost solenoid (-)	w108_c
19	Throttle pos 1 main	w128_c
20	Throttle pos 2 sub	w129_c
21	Injector 1 drive (-)	w108_c
22	Injector 2 drive (-)	w109_c
23	Injector 3 drive (-)	w110_c
24	Injector 4 drive (-)	w111_c
25	+5 V coolant press	w_5v_brail_c47
26	Coolant press signal	w35_c
27	+5 V oil pressure	w_5v_brail_c45
28	Oil pressure signal	w35_c
29	Gnd oil/crit/temps	w_gnd_b_c19
30	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	w_rail_sp_sw12_c
31	Flex fuel signal	w48_c
33	Intake temp signal	w202_c
34	Gnd IAT/TPS/coolant	w_rail_sp_gndout_c
41	+5 V throttle sensor	w_rail_sp_5v_c
42	Throttle pos 1 main	w128_c
44	Throttle pos 2 sub	w129_c
45	ETB motor +	w124_c
46	ETB motor -	w215_c
51	+5 V turbo speed	w_5v_a_c40
52	Turbo speed signal	w16_c
53	Gnd turbo speed	w_gnd_a_c43
56	Boost solenoid (-)	w103_c
58	Charge temp signal	w204_c
59	Oil temp signal	w203_c
61	+8 V to cam sensor	w_rail_sp_8v_c
63	Cam sensor signal	w_cam_sig_c
64	Gnd cam sensor	w_gnd_a_c42
65	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	w_rail_sw12_b_c_ac
67	Gearbox speed signal	w_vss_sig_c
68	Coolant temp signal	w201_c
70	Knock sensor signal	w205_c
71	Knock sensor (-)	w12_c
72	Crank sensor signal	w200_c
74	Gnd crank sensor	w_gnd_a_c41
75	Back-up sw +12 V in	w_rev_feed_c
77	Reverse sw +12 V out	w_rev_sw_c
78	CAN H to CAN-Lambda	wc_h_bh_bc
79	CAN L to CAN-Lambda	wc_l_bh_bc
80	EFI relay k_efi -> LOGIC MRS 7.5 A fuse	w_fb_mrs_en
83	EHPS pump speed	w_mrs_c
84	EHPS relay coil (-)	w_eps_rly
85	EFI relay k_efi -> LOGIC LAM 7.5 A fuse	w_lam12_cp
95	CAN-Lambda ground	w_lamg_ac

CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses

ECU-A - Link G4X plug A (centre dash)
6-1437290-1

9	Cam sensor signal	w_inj2_pin
18	Pedal 1 signal main	w38
19	A/C cancel (ACT)	w_ac_kill
19	Fuel pump relay (-)	w_fp85
20	ETB relay coil (-)	w81
23	FL conditioned out	cab_hpur_l
26	Start relay coil (-)	w_str85
28	EFI relay coil (-)	w76
29	Rad fan relay (-)	w_fan85

ECU-B (Link G4X plug B, centre dash)
6-1437290-3

5	ETB motor +12 V	w82_ac
10	A/C clutch relay (-)	w_acc_trig
12	EPS enable relay (-)	w_eps_trig
13	Cond fan relay (-)	w_fan285
19	Rear right wheel spd	cab_rspur_r
20	Rear left wheel spd	cab_rspur_l
21	FR conditioned out	cab_hpur_r_ac
24	Fuel level signal	w_ff_sig_pin
27	Anti-theft input	w_at_immob2
28	Ignition sw sense	w_ign
30	Start request	w_start
31	A/C request button	w_ac_btn
32	Evap temp signal	w_evap_temp
33	Pedal 2 signal sub	w_aps_vpa2

SP-AC-PSW - At A/C amp - SP-AC-PSW: A/C pressure switch...
solder splice (node)

-	A/C pressure sw out	w_ac_psw_in
-	A/C pressure sw out	w_ac_psw_sw
+	A/C pressure sw (+)	w_acc86

SP-REV - reverse switch out to CSB3 opto + reverse relay
solder splice (node)

-	Reverse relay coil+	w_rev_coil_c
-	Reverse to opto in 2	w_sw7_c

R-AC-BTN-HI - 10k A/C switch divider, top
inline part

1	A/C switch V-G in	w_ac_btn_in
2	A/C btn divider mid	w_ac_btn_hi

R-AC-BTN-LO - 3.9k A/C switch divider, bottom
inline part

1	A/C btn divider mid	w_ac_btn_lo

R-CAM-PULLUP - Cam pull-up 1.8 k to +8 V
inline part

2	Cam pull-up 1.8k	w_cam_pullup_sig

R-EVAP - 2.2k evap thermistor pull-up to +5V
inline part

1	Evap pull-up 2.2k	w_evap_pu

R-FUELLVL - 470R fuel level pull-up to +5V
inline part

1	Fuel lvl pull-up	w43

IX-FUEL-P - IX-FUEL inline to fuel sender lead (trunk end)
DT04-2P

1	Fuel level signal	w_ff_sig

T-EVAP-TEMP - OEM evaporator thermistor A24 pin 1, at the ...
eyelet

-	Evap thermistor sig	w_evap_sig

OEM ST1 lead B-W at old R/B No.4 pin 1, RH kick panel (on ...
factory lead tap

1	Start sw ST1 feed	w_start_in_cp

OEM A/C amplifier
factory lead tap

11	A/C cancel (ACT)	w_ac_kill

K-FP - Fuel Pump Relay, TE HCR150 (seated in the battery relay b...
925710-K000-B100

85	Fuel pump relay (-)	w_fp85

K-STR - Start Relay (seated in the POWER box R3) - ACTIVE-LOW...
6-1414141-3

85	Start relay coil (-)	w_str85

K-EFI - EFI Main Relay (seated in the POWER box R2) - ACTIVE-L...
6-1414141-3

85	EFI relay coil (-)	w76

K-FAN - Radiator Fan Relay (on/off) (seated in the POWER box R...
6-1414141-3

85	Rad fan relay (-)	w_fan85

K-ETB - ETB Power Relay (seated in the LOGIC box M4) - ACTIVE...
6-1393216-3

85	ETB relay coil (-)	w81
87	ETB motor +12 V	w82_ac

IX-WS-F-S - to front VR conditioner
DT06-6S

3	FL conditioned out	cab_hpur_l
4	FR conditioned out	cab_hpur_r_ac

APS-PEDAL - accelerator pedal sensor
solder splice (node)

3	Pedal 2 signal sub	w_aps_vpa2
6	Pedal 1 signal main	w38

K-EPS-EN - EPS Enable Relay (seated in the LOGIC box M6) - AC...
6-1414141-3

85	EPS enable relay (-)	w_eps_trig

K-FAN2 - Condenser Fan Relay (on/off) (seated in the POWER bo...
6-1393216-3

85	Cond fan relay (-)	w_fan285

IX-WS-R-S - IX-WS-R inline to rear VRC (dash above ECU)
DT06-6S

3	Rear left wheel spd	cab_rspur_l
4	Rear right wheel spd	cab_rspur_r

ALARM - Alarm module - main harness (DT06-12SA) (on ST185-A...
DT06-12SA

9	Anti-theft input	w_at_immob2

OEM J/B No.1 (LH kick panel) 1H pin 7, B-O
factory lead tap

7	Ignition sw sense	w_ign

K-ACC - A/C Clutch Relay (on/off) (seated in the LOGIC box M5) - ...
6-1414141-4

85	A/C clutch relay (-)	w_acc_trig
86	A/C clutch relay (+)	w_acc86

SP-CAM - At the ECU plugs - SP-CAM: cam signal + 1.8k pull-up t...
solder splice (node)

-	Cam sensor signal	w_inj2_pin
-	Cam pull-up 1.8k	w_cam_pullup_sig

OEM A5, factory Y-R lead
factory lead tap

4	A/C pressure sw out	w_ac_psw_in

OEM A/C amplifier
factory lead tap

13	A/C pressure sw out	w_ac_psw_sw

K-REV - Reverse Relay (seated in the LOGIC box M7) - coil+ (86) f...
6-1414141-4

86	Reverse relay coil+	w_rev_coil_c

OPTO-B - Optocoupler board, 8 channel (Dan's, generic EL814) - i...
eyelet

IN2	Reverse to opto in 2	w_sw7_c

SP-AC-BTN - A/C switch divider mid-point to ECU-B B31
solder splice (node)

-	A/C request button	w_ac_btn
-	A/C btn divider mid	w_ac_btn_hi
-	A/C btn divider mid	w_ac_btn_hi

T-AC-BTN - OEM dash A/C switch V-G lead (old A15 pin 3)
eyelet

+	A/C switch V-G in	w_ac_btn_in

LOGIC box - custom fuse + relay box, glove box inboard (cad/fus...
925710-K000-0000 · loom relay box

BUS2-13	Start request	w_start
BUS2-...	Start sw ST1 feed	w_start_in_cp

SP-FUEL-LVL - fuel sender signal + 470R pull-up to ECU-B B24
solder splice (node)

-	Fuel level signal	w_ff_sig
-	Fuel level signal	w_ff_sig_pin
-	Fuel lvl pull-up	w43

SP-EVAP - evap thermistor + 2.2k pull-up to ECU-B B32
solder splice (node)

-	Evap temp signal	w_evap_temp
-	Evap pull-up 2.2k	w_evap_pu
-	Evap thermistor sig	w_evap_sig

ECU SPLICE POINT - power, grounds and sensor rails (bussed DT plugs, soldered splices, NEG bar)

ECU-A - Link G4X plug A (centre dash)
6-1437290-1

5	EFI relay k_efi -> LOGIC SW12 7.5 A fuse...	w_efi_pin
6	+8 V sensor supply	w19
24	Sensor ground	w68
25	ECU power ground	w70
32	+5 V sensor supply	w67
34	ECU power ground	w_a34

ECU-B (Link G4X plug B, centre dash)
6-1437290-3

22	Sensor ground B	w69
25	ECU power ground	w_b25_ac
34	ECU power ground	w_b34_ac

BUS-5V-2 - DT06-8 plug (ECU splice point)
DT06-08SA

1	+5 V from BUS-5V	w_5v_bus_link
2	+5 V cruise ladder	w_cru_5v_ac
3	+5 V evap pull-up	w_evap_5v
4	+5 V fuel lvl pullup	w44

BUS-SW12 - SW12 bus plug DT06-12 (glove box, by LOGIC b...
DT06-12SA

2	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	w_fb_sw12

SP-A5 - ECU splice point - SP-A5: ECU main feed (SW12 + IG...
solder splice (node)

-	EFI relay k_efi -> LOGIC SW12 7.5 A fuse...	w78
-	EFI relay k_efi -> LOGIC SW12 7.5 A fuse...	w_ig2_diode
-	EFI relay k_efi -> LOGIC SW12 7.5 A fuse...	w_a5_pin

R-AC-BTN-LO - 3.9k A/C switch divider, bottom
inline part

2	A/C btn divider gnd	w_ac_btn_gnd

R-CAM-PULLUP - Cam pull-up 1.8 k to +8 V
inline part

1	+8 V to cam pull-up	w_cam_pullup_8v

APS-PEDAL - accelerator pedal sensor

1	+5 V pedal sensor 2	w210
2	Pedal 2 ground	w_aps_gnd2
4	+5 V pedal sensor 1	w_aps_vc1
5	Pedal 1 ground	w40

IX-FUEL-P - IX-FUEL inline to fuel sender lead (trunk end)
DT04-2P

2	Fuel level ground	w45

IX-WS-F-S - to front VR conditioner
DT06-6S

1	VRC-F +5V	cab_hpur_5v
2	VRC-F GND	cab_hpur_gnd

IX-WS-R-S - IX-WS-R inline to rear VRC (dash above ECU)
DT06-6S

1	VRC-R +5V	cab_rspur_5v
2	VRC-R GND	cab_rspur_gnd

POWER-NEG-BAR - POWER box NEG bar M8 stud (glove box)
stud

-	CSB3 + opto ground	w_csbg_ac
-	ECU power ground	w66

T-EVAP-RTN - OEM evaporator thermistor A24 pin 2, at the e...
eyelet

-	Evap thermistor gnd	w_evap_rtn

OEM J/B No.1 (LH kick panel) 1H pin 7, B-O
factory lead tap

7	IG2 wake feed	w_ig2_in

R-CRUISE - resistor r_cruise on CSB3
inline part

1	+5 V cruise ladder	w_cru_5v_ac

R-EVAP - 2.2k evap thermistor pull-up to +5V
inline part

2	+5 V evap pull-up	w_evap_5v

R-FUELLVL - 470R fuel level pull-up to +5V
inline part

2	+5 V fuel lvl pullup	w44

SP-GND-ECU - ECU splice point - SP-GND-ECU: ECU power grou...
solder splice (node)

-	ECU power ground	w70
-	ECU power ground	w_a34
-	ECU power ground	w_b25_ac
-	ECU power ground	w_b34_ac
-	ECU power ground	w66

SP-8V - ECU splice point - SP-8V: +8 V cam sensor supply (ECU ...
solder splice (node)

-	+8 V to cam pull-up	w_cam_pullup_8v
-	+8 V sensor supply	w19

SP-SW12 - ECU splice point - SP-SW12: switched +12 V (LOGIC S...
solder splice (node)

-	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	w_fb_sw12
-	EFI relay k_efi -> LOGIC SW12 7.5 A fuse (+ l...	w78

BUS-GNDOUT-A - DT06-12 plug (ECU splice point)
DT06-12SA

1	Sensor gnd from A24	w68
7	Pedal 1 ground	w40
8	Pedal 2 ground	w_aps_gnd2
9	Front VRC ground	cab_hpur_gnd

BUS-5V - DT06-12 plug (ECU splice point)
DT06-12SA

1	+5 V from ECU A32	w67
8	+5 V pedal sensor 1	w_aps_vc1
9	+5 V pedal sensor 2	w210
10	+5 V front VRC	cab_hpur_5v
11	+5 V rear VRC	cab_rspur_5v
12	+5 V to BUS-5V-2	w_5v_bus_link

BUS-GNDOUT-B - DT06-12 plug (ECU splice point)
DT06-12SA

1	Sensor gnd from B22	w69
2	Fuel level ground	w45
3	Rear VRC ground	cab_rspur_gnd
4	A/C btn divider gnd	w_ac_btn_gnd
5	Evap thermistor gnd	w_evap_rtn

D-IG2-WAKE - IG2 wake diode (1N5822)
inline part

1	IG2 wake feed	w_ig2_in
2	EFI relay k_efi -> LOGIC SW12 7.5 A fuse (+ l...	w_ig2_diode

SP-CSB-GND - CSB3 ground splice - CSB3 pin 2, opto board CO...
solder splice (node)

-	CSB3 + opto ground	w_csbg_ac

SHIELD DRAINS - screens to A7 / B17 (R207), CAN screen to the CAN trunk

ECU-A - Link G4X plug A (centre dash)
6-1437290-1

7	Shield drain	w_drain_ecu_a

ECU-B (Link G4X plug B, centre dash)
6-1437290-3

17	Shield drain B	w_drain_ecu_b

DRB102-CABIN - DRB102 receptacle (RH firewall)
DRB12-1025NE-4169

32	Flex fuel shield	w_shc_flex
47	Throttle shield	w_shc_etbs
50	ETB motor shield	w_shc_etbm
57	Turbo speed shield	w_shc_turbo
66	Cam sensor shield	w_shc_cam
69	Gearbox speed shield	w_shc_vss
73	Knock sensor shield	w_shc_knock1
76	Crank sensor shield	w_shc_crank
81	CAN-Lambda shield	cab_canbh_bc_sh_screen

IX-WS-F-S - to front VR conditioner
DT06-6S

5	WS-F screen drain	cab_hpur_sh

IX-WS-R-S - IX-WS-R inline to rear VRC (dash above ECU)
DT06-6S

5	WS-R screen drain	cab_rspur_sh

CAB-APS1 cable screen (floats at the device end)
cable screen

-	Pedal 1 cable shield	cab_aps1_sh

CAB-APS2 cable screen (floats at the device end)
cable screen

-	Pedal 2 cable shield	cab_aps_sh

CAB-FSPUR-FR-AC cable screen (floats at the device end)
cable screen

-	FR wheel spd shield	cab_hpur_fr_sh_ac

SP-SHIELD-A - 6 mm behind the ECU-A plug - SP-SHIELD-A: drai...
solder splice (node)

-	Shield drain	w_drain_ecu_a
-	Shield drain	w_shd_a_aps
-	Shield drain	w_shd_a_wsf
-	Crank sensor shield	w_shc_crank
-	Cam sensor shield	w_shc_cam
-	Turbo speed shield	w_shc_turbo
-	Flex fuel shield	w_shc_flex
-	Throttle shield	w_shc_etbs

SP-SHIELD-B - 6 mm behind the ECU-B plug - SP-SHIELD-B: drai...
solder splice (node)

-	Knock sensor shield	w_shc_knock1
-	ETB motor shield	w_shc_etbm
-	Gearbox speed shield	w_shc_vss
-	Shield drain B	w_drain_ecu_b
-	Shield drain B	w_shd_b_wsr

SP-CAN-SH-BH - CAN trunk, bulkhead B branch - SP-CAN-SH-BH...
solder splice (node)

+	CAN-Lambda shield	cab_canbh_bc_sh

SP-SHIELD-B-WSR - At the rear WS inline - SP-SHIELD-B-WSR: r...
solder splice (node)

-	WS-R screen drain	cab_rspur_sh
-	Shield drain B	w_shd_b_wsr

SP-SHIELD-A-WSF - At the front WS inline - SP-SHIELD-A-WSF: fr...
solder splice (node)

-	Shield drain	w_shd_a_wsf
-	WS-F screen drain	cab_hpur_sh
-	FR wheel spd shield	cab_hpur_fr_sh_ac

SP-SHIELD-A-APS - At the pedal branch - SP-SHIELD-A-APS: ped...
solder splice (node)

-	Shield drain	w_shd_a_aps
-	Pedal 1 cable shield	cab_aps1_sh
-	Pedal 2 cable shield	cab_aps_sh

REVISIONS

REV	DATE	BY	DESCRIPTION
A	2026-10-19	DG	First issue: own drawings from the wire list (no harness.design app); loom A + B as one harness (...

ST185-AB-cabin - ECU harness (loom AB), cabin side
Schematic - every wire, grouped by job

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-AB-cabin-SCH	A	1 OF 1	889 x 686 mm	DG	2026-10-19

ST185 Celica (living-relation) · generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale